

Unmasking Public Sentiment: A Sample Efficient Approach to Analyzing Twitter Opinion on U.S. Aid to Ukraine

Satanu Ghosh
University of New Hampshire
satanu.ghosh@unh.edu

Souvick Ghosh
San José State University
souvick.ghosh@sjsu.edu

Nikhil Dewitt
University of California Los Angeles
nikhildewitt@g.ucla.edu

Denise McCoy
San José State University
denise.mccoy@sjsu.edu

Abstract

The U.S. aid to Ukraine is a bipartisan topic of extreme socio-political importance. While several organizations have conducted surveys to understand the public stance on this topic, there is no research to date that analyses public opinion on social media, possibly due to the lack of annotated data. Therefore, this research compares several sample-efficient methods (including in-context learning) to analyze tweet sentiments with minimal training data. First, we collect 11,289 tweets about the U.S. aid to Ukraine and mapped them to U.S. states. Next, we explore three different approaches to sentiment analysis: tool-based, embedding-based, and prompt-based. Our results indicate that GPT-4 Few Shot improves accuracy by 121.8% and 77.5% over TextBlob and Vader, respectively. Our geospatial analysis shows that Indiana has the most negative normalized net sentiment (NNS) of -0.83, while Vermont has the most positive NNS of +0.33. Finally, we perform a detailed thematic analysis to identify the common arguments that support or oppose the aid. We highlight that our results do not correlate with media surveys, possibly due to the presence of echo chambers and algorithmic biases.

Keywords: Social Media, Twitter Data Analysis, Few-shot Learning, Zero-shot Learning, Sentiment Analysis

1. Introduction

The Ukraine-Russian conflict has created a unique geopolitical tension in the United States. The U.S. views the location of Ukraine as the key to maintaining peace in Eastern Europe and has supported democratic

elections and aspirations of Ukraine, since 2014¹. However, building international influence comes at a significant cost. Since the start of the war in Ukraine in February 2022, the U.S. has sent significant monetary aid to Ukraine to thwart the Russian attempts at invasion. Providing such aid not only strengthens democratic institutions in Ukraine but also supports humanitarian efforts to provide food, shelter, and medical assistance to those displaced in the conflict.

When we look at public opinion within the United States, we see that it presents an interesting picture. While some sections of the U.S. population believe in countering Russian aggression in Eastern Europe, others question the necessity of providing aid to a foreign country, arguing that resources should be prioritized for domestic needs. In the initial phases of the war, support for military aid to Ukraine was largely bipartisan, as 81% of Democrats and 71% of Republicans supported the U.S. aid policy. However, towards the end 2022, attitudes began to shift, particularly with Republicans, as support for sending aid dropped from 80% in March to 55% in December². Overall, Democrats and President Biden have been more supportive of aid to Ukraine, while the Republican Party has shown skepticism toward spending millions of dollars to sponsor military equipment for Ukraine.

Considering the polarization around the topic of U.S. aid to Ukraine, and the partisan differences and the differential portrayal of the situation by various media outlets, it is timely to assess the public opinion on the situation, which could be particularly important for the Presidential election in 2024. Since the U.S. follow a federal system of governance with power divided

¹<https://ua.usembassy.gov/fact-sheet-u-s-assistance-ukraine-since-february-2014/>

²<https://www.pewresearch.org/politics/2022/03/15/public-express-es-mixed-views-of-u-s-response-to-russias-invasion-of-ukraine/>

into national and individual state governments, it is also important to consider the sentiment expressed by the different states. As of 2023³, Twitter has the largest representation from the U.S. (80 million) and is predominantly male-dominated (68.1%). Therefore, we lean on Twitter to assess the public opinion on the topic. While social media posts are a representation of the broader national sentiment, the algorithmic recommendations (the tweets that show up on the personalized “For You” feed) are not always unbiased and often lead to echo chambers where social media users subscribe to and interact with only content which supports their particular belief system. Therefore, any attempt at deciphering public opinion on social media should be approached with a degree of scientific skepticism and a mixture of scientific approaches.

Another major challenge of performing such sentiment analysis is the lack of labeled training data (Birjali et al., 2021; Kanojia and Joshi, 2023). With massive volumes of data generated on social media platforms, it is expensive and time-consuming to manually assess the sentiment polarity and develop gold-standard data (Aqlan et al., 2019). However, it is difficult to develop accurate prediction models without adequate training data. The performance of most off-the-shelf sentiment classifiers are poor (Ghosh and Thajudeen, 2023), and most research approaches rely on annotating the data first and then performing analyses, which delays the insights obtained. A faster turnaround time could be significantly useful for emergency responses (e.g., COVID-19 pandemic), natural disasters (e.g., California wildfires), or other important events (e.g., Presidential elections).

In this work, we consider the following research questions:

RQ1: For the collected tweets, how does public opinion vary across the U.S. states?

RQ2: What are the most common reasons for supporting or opposing U.S. aid to Ukraine?

RQ3: Is our data reflective of the common perception around support for U.S. aid to Ukraine?

We hypothesize that a multifaceted analysis of public opinion data can yield more comprehensive insights. The geographic analysis from RQ1 and the thematic understanding from RQ2 are crucial to this approach. However, recognizing that results can be influenced by inherent biases in the data source, RQ3 aims to validate our findings by comparing social media opinions with more reliable public polls. This comparison helps draw more accurate inferences about national public opinion and elucidate the unique characteristics of social media data/platforms. Furthermore, it paves the way for future

research on developing de-biasing methods for social media-derived datasets.

To answer our RQs, we collected tweets on the topic of U.S.-Ukraine aid (see Section 3.1) and geotagged the collected tweets using location information. Then we used sample-efficient approaches like Zero-Shot Learning (ZSL) and Few Shot Learning (FSL) (with five training examples) to predict the sentiment polarities for the tweets. We compare the ZSL and FSL predictions to traditional machine learning classifiers using an embedding-based approach. The most accurate sentiment prediction model was used to explore how public opinion is divided across different states. Next, we perform a deep dive into the dataset and identify the most common themes for the public to support or oppose the U.S. aid to Ukraine. Lastly, we explore the literature in relation to the ways in which social media recommender systems interact with human behavior to assess why our data might support or differ from the perceived dominant narrative viewpoint of U.S. aid to Ukraine.

The rest of the paper is organized as follows: In section 2, we discuss some of the related research in social media and text mining, geotagging, and generative large language models (LLMs). We cover the psycho-social structures of social media spaces and the current state of recommender systems. Section 3 explains our methodology, including data collection and augmentation. We present our results in section 4, and answer the research questions in section 5. Finally, we conclude the paper and discuss limitations and future research directions in sections 6 and 7, respectively.

2. Background

To reach our research objective, it was important to survey some pertinent research papers on social media and analysis.

2.1. Social Media Analytics

Social Media as a Public Sphere: Social media have redefined the idea of public sphere (Kruse et al., 2018) in today’s age of information. The discourse taking place on social media platforms help researchers to explore socially relevant topics and measure public opinions on topics like the COVID-19 pandemic (Jalil, 2022; Lanyi, 2022), political discourse (Noor and Turetken, 2023; Taillon and Mentzer, 2023), and government decisions (Mane et al., 2022). Noor and Turetken (2023) investigated the keywords that drive the sentiment polarity in social media, while Taillon and Mentzer (2023) explored the effect of gender and political ideology on political sentiment. Several

³<https://www.searchlogistics.com/learn/statistics/twitter-user-statistics/>

researchers (V. Kumar, 2022; Vahdat-Nejad, 2023) have also performed geospatial analysis to understand the variations of human perspective based on demographics. **Text Analytics:** Text analytics is a common Natural Language Processing (NLP) task, and social media text – which is often more faceted than regular text (S. Kumar et al., 2012) – has become a popular topic of investigation in recent years. Opinion mining is a specific case of text analytics where the model is trained to predict the opinion expressed in the text. However, training a model is a data-hungry process that requires massive amounts of human-annotated data, and is, therefore, costly. For example, Jalil (2022) used 45,000 tweets annotated with sentiment labels for training. The alternative to annotating data is to use off-the-shelf models (Al-Natour and Turetken, 2020; Elmitwalli and Mehegan, 2024; Lanyi, 2022) or publicly available annotated datasets on similar topics (Mane et al., 2022). However, the annotation labels from most off-the-shelf models lack specificity and need retroactive adaptation.

Large Language Models (LLMs) as Knowledge Bases: Pre-trained LLMs are considered to contain tacit knowledge (Petroni, 2019) as they are trained in an unsupervised fashion on massive text datasets. For example, BERT is a transformer-based model that is trained on English Wikipedia. The embeddings generated by these models perform very well for tasks like clustering, classification, and ranked retrieval (Keraghel et al., 2024). More recent LLMs, like GPT-3 (Brown et al., 2020), are pre-trained on Common Crawl, Book Corpus 1, Book Corpus 2, and Wikipedia, enabling them to assimilate the latent structures and embedded knowledge on various topics. These developments have enabled generative LLMs to follow instructions and work effectively in a more sample-efficient setting (Brown et al., 2020). Subsequently, the rise of prompt engineering and the concept of zero-shot and few-shot in-context learning have been effectively used to classify tweets and other social media data (Mu, 2023) and analyze sentiments (Yadav and Chakraborty, 2021). One clear advantage of using generative LLMs is their adaptability to unseen annotation labels, along with their probabilistic understanding of natural language patterns.

2.2. Social Media Recommender Systems & Echo Chambers

Recommender System: There is an overabundance of data on social media platforms, and users often find it overwhelming to find information (Rodriguez et al., 2014). As a solution, social media platforms provide

a personalized experience to users by recommending information that is more correlated to their preference (Li, 2024), leading to more engagement. The recommender system used in Twitter is a three-part system⁴: a knowledge graph, a ranking system, and a heuristic-based filtering method (Satuluri, 2020). It is important to note that there is a lack of transparency to the inner workings of these models as they are supported by automatically created graph networks, unsupervised neural clustering and ranking models (that are black boxes), and filtering rules set by the platform developers. **Echo Chambers:** Algorithmic recommendations are prone to biases, thereby creating echo chambers, where a group of users subscribes to content that fits their belief system (Cinelli, 2021). These users exclude and discredit others, therefore exacerbating political polarization and promoting the spread of misinformation (Alatawi et al., 2021). Echo chambers are caused and exacerbated by two areas of biases: 1) internal biases and homophily, and 2) external biases, such as biased recommender systems and exposure to biased communities. Cinelli (2021) demonstrated the existence of echo chambers for several topics on multiple social media sites, including Twitter/X. In such spaces where there is a lack of diverse content, there is an increase in radical thoughts and behaviors from the users, highlighting the need for diverse media networks (Atari et al., 2022). Tweets containing morally and emotionally volatile content are found to become more popular (Brady et al., 2020) within an echo chamber as they are more engaging (and algorithms are optimized for higher engagement). It is also found that negative language content spreads further and faster in echo chambers (Antypas et al., 2023; Schöne et al., 2021) in contrast to positive or neutral content.

3. Methodology

To analyze the public opinion on Twitter/X on the U.S. aid to Ukraine, we employ a mixture of methods which we explain in the following subsections.

3.1. Data Collection and Preparation

While there are some social-media datasets on the Ukraine-Russia conflict (Aslan, 2023; Chen and Ferrara, 2023), they were not related to the U.S. aid to Ukraine or the U.S. demographic. Therefore, we used Snsrape⁵ to scrape tweets related to the Ukraine Crisis for a period ranging from February 1, 2022, to December 21, 2022. To target tweets within the United States,

⁴<https://blog.x.com/engineering/en-us/topics/open-source/2023/twitter-recommendation-algorithm>

⁵<https://github.com/JustAnotherArchivist/snsrape>

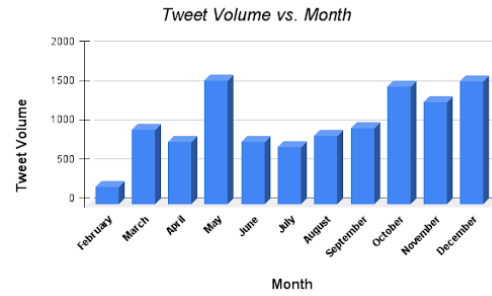
the geographic center of the continental United States, Lebanon K.S., was used with a radius of 10,000 km. A daily target of 1000 tweets was set, and the keyword ‘Ukraine’ was used for the search. Later, to select the tweets pertaining to debates over U.S. aid to Ukraine, we filtered tweets using the following keywords: ‘aid,’ ‘40 billion,’ ‘money,’ ‘spending,’ and ‘billion.’ After filtering for geolocation and aid-related tweets, our initial dataset contained 11,289 tweets. Next, we used Shapely⁶ to determine the state where the tweet originated from and added the information to the dataset.

We randomly selected 700 instances from the entire collection and split them into three parts: train, validation, and test sets. These tweets were annotated by human annotators. To replicate a data-scarce situation, our train and validation set only contained 100 tweets. The test set contained 424 tweets (76 tweets were removed as annotators could not agree on the sentiment labels). As seen in Figure 1a, the largest volume of tweets in our dataset was from May and December, followed by October and November. In late October and November, the Defense Department announced that it would provide security assistance to Ukraine, which led to increased social media activity during these months. Figure 1b shows the tweet volume by states in our dataset. Florida, Texas, and California show the maximum number of tweets, with 10.1%, 9.1%, and 7.3%, respectively, of total tweets.

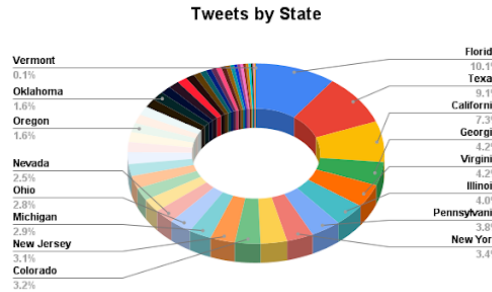
Geo-locating Tweets Once we collected the tweets, we used the GeoPandas⁷ and a GeoJSON file to match the coordinates in the tweet to a U.S. state. Many of the tweets listed a state in the location attribute, but we used the coordinates since it is more accurate and reliable. Tweets in the continental U.S. were labeled with their state, and the tweets from outside the United States were removed from the dataset. Next, exploratory data analysis was used to determine the volume and relative volume of the tweets per state, calculated as a ratio of volume over state population. The GeoPandas library was used to convert the GeoJSON into individual polygons, and the MapClassify library was used to create maps for each state. The Plotly-Express Library, particularly the Mapbox Feature, was used to create the bubble plot for sentiment by state. It should be noted that the accuracy of geolocation data depends on whether the user has allowed sharing precise location data. If precise location sharing is disabled, platforms rely on IP-based geolocation, which might not be fully accurate since IP addresses can be manipulated using VPNs and other location-spoofing tools.

⁶<https://pypi.org/project/shapely/>

⁷<https://geopandas.org/en/stable/>



(a) Tweet Volume by Month.



(b) Distribution of Tweets by State.

Figure 1: Data Collection and Geo-tagging.

3.2. Sentiment Analysis

While world events influence the topics trending on social media, public opinion varies depending on the demographics of the posters and the consumers. To analyze data and train prediction models, researchers different annotation labels with varying levels of granularity. However, since the manual annotation process is time-consuming and expensive, researchers rely on off-the-shelf classifiers. Recently, with the inception of pre-trained language models, some other methods have emerged, like word-embedding-based classification and in-context classification. For this task, we investigated which approach would perform best in a **data-scarce setting**. We tested three different types of classification strategies: tool-based, embedding-based, and prompt-based.

Tool-based Approach: We used two off-the-shelf classification tools – TextBlob⁸ and Vader (Hutto and Gilbert, 2014) – which are widely used for sentiment analysis (SA) and opinion mining tasks on social media data (Al-Natour and Turetken, 2020; Elmitwalli and Mehegan, 2024). Both tools use heuristics to determine the sentiment of any given text and have been extensively validated in prior works. While VADER uses a rule-based approach, TextBlob is a hybrid tool

⁸<https://textblob.readthedocs.io/en/dev/>

using lexicon-based techniques and machine learning. These tools are trained on SA datasets with some fixed labeling schema. While researchers do not need to provide annotated data to these tools, the tools come with an inherent limitation: the labels are not customizable in granularity and may not capture the complexity and context of the text adequately.

Embedding-based Approach: BERT is a pre-trained model that can be used to produce representation for classification purposes (CLS embedding). Leveraging this feature, we obtained a 256-dimension embedding of the tweets where the average length of tweets was 175 words. We trained multiple machine learning classifiers – Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Random Forest and Logistic Regression – with BERT-embeddings of 100 tweets.

Prompt-based Approach: Large language models like GPTs are capable of following human instruction, as they are trained on various tasks grounded in language. Being generative in nature and following Markovian assumptions, these models are very sensitive to the instruction or prompt. An advantage of using GPT-like models is that they do not require much training data (if any at all) to produce outputs. For our task, we tried two different types of prompting: zero-shot and few-shot in-context learning with GPT-3.5-turbo and GPT-4-0613. Zero-shot prompting only instructs the model to do a task, whereas few-shot prompting also provides a few exemplars to the model to show how the task should be done. A batch of 10-tweets was passed to the model for classification. The same instruction was provided to the model in both zero-shot and few-shot training. For the few-shot, we passed 5 exemplars picked from the training collection (100 tweets), of which the “supports” and “irrelevant” labels had two instances each, and “opposes” had one instance.

3.3. Qualitative Analysis of Themes

While thematic analysis has its roots in psychological research (Braun and Clarke, 2006), it has gained popularity for social media analysis (Xiong et al., 2019). We conducted a qualitative, reflective thematic analysis of a randomly selected sample of 100 tweets (N=100) to identify the major themes opposing and in support of the U.S. aid to Ukraine. Two authors reviewed the tweets sample to develop independent codes associated with the tweets. Next, they compared notes and themes, ultimately selecting five major themes supporting and opposing the aid (reported in Section 5.2).

Table 1: Classification result on validation set.

Model Name	Accuracy	Precision	Recall
<i>Tool-based Approach</i>			
TextBlob	0.32	0.47	0.32
Vader	0.40	0.48	0.40
<i>Embedding-based Approach</i>			
BERT Embeddings + SVM	0.58	0.34	0.43
BERT Embeddings + KNN	0.58	0.34	0.58
BERT Embeddings + Random Forest	0.59	0.63	0.59
BERT Embeddings + Logistic Regression	0.60	0.68	0.60
<i>Prompt-based Approach</i>			
GPT-3.5 Zero Shot	0.61	0.57	0.61
GPT-3.5 Few Shot	0.60	0.59	0.60
GPT-4 Zero Shot	0.67	0.67	0.67
GPT-4 Few Shot ^Δ	0.71	0.73	0.71

4. Results & Analysis

In this section, we discuss the results of the different methods on our validation dataset and perform a detailed error analysis for the best-performing model on our test data.

Validation Results: The results of the sentiment classification approaches are reported in Table 1. Since our dataset contained an unbalanced class representation, all the reported metrics are weighted averages. GPT-4 Few Shots performed best in terms of accuracy, precision, and recall. Although using BERT embeddings as features for traditional machine learning algorithms is a fast and inexpensive way to analyze textual data, yet it was not very effective in a limited data scenario (with small gold standard annotations). However, it is important to note that BERT embeddings and Logistic Regression outperform GPT-3.5 Zero Shot. Other off-the-shelf sentiment classification tools like Vader and TextBlob, popular among social science and information science researchers, were also used to establish a baseline. However, their performance was much inferior compared to the other models.

Test Prediction and Error Analysis: As our validation set was quite small (N=100) and the generation process of GPT-4 is non-deterministic in nature, we wanted to verify the generalizability of results for our best-performing model (GPT-4 Few Shot). GPT-4 Few Shot achieved a prediction accuracy of 76.2%, precision of 78%, and recall of 76.2%, all scores calculated as a weighted average of the three classes. A detailed confusion matrix has been presented in Figure 2. This performance is slightly better than that achieved by the model for our validation set. Therefore, we can conclude that for this particular setting (the prompt, the labels, and the problem), the performance of GPT-4 Few Shot is generalizable to larger samples.

From the confusion matrix, we can see four major

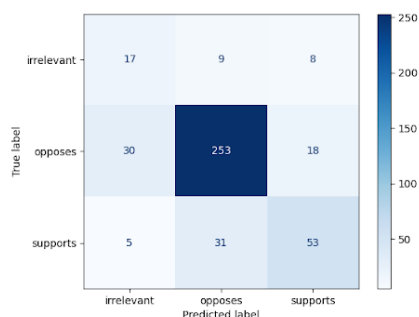


Figure 2: Confusion Matrix for GPT-4 Few Shot for Testset (N=424).

error types caused by *Double Negation* or *Critique of a Related Entity*, *Lack of Semantic Correlation*, *Use of Satire*, and *Lack of Specificity*. Examples of the errors can be found in Table 2.

(1) **“supports” predicted as “opposes”:** 31 tweets that supported U.S. aid to Ukraine were predicted to be “opposes.” Upon further investigation, we found that most of these tweets did not explicitly mention support for the aid. The tweets were criticizing people or entities who are against financial aid to Ukraine (*Double Negation*). We also found that “aid” was not the only topic of interest in these tweets; other societal issues like social security, medicare, and reproductive rights were also mentioned (*Critique of a Related Entity*).

(2) **“opposes” predicted as “irrelevant”:** About 30 tweets that were annotated as “opposes” were erroneously predicted as “irrelevant.” Upon further checking, we found that most of these errors pertain to Natural Language Understanding. For example, the words “fix” or “laundry”, when used independently, are not derogatory or negative, but their usage in the context of the aid hints at corruption. GPT-4 did not understand the subtle underlying nuances of human language, which led to errors (*Lack of Semantic Correlation*).

(3) **“opposes” predicted as “supports”:** Among the 18 tweets belonging to this error category, most were satirical in tone (*Use of Satire*).

(4) **“irrelevant” predicted as “opposes” or “support”:** Tweets that were irrelevant to aid were mostly personal attacks on individual political personalities. These tweets mentioned the aid in the passing or as facts related to the aid. The predictions made by the model were incorrect because the model considered the overall sentiment of the tweet instead of the particular context (the U.S. aid to Ukraine) (*Lack of Specificity*).

Table 2: Common Classification Errors.

Error Type	Example
Critique of a Related Entity or Double Negation	Folks, do not let Republicans take over. They would end Democracy, Cut Social Security, Cut Medicare, End Reproductive Rights, Impeach Biden, Defund the DOJ & FBI, Cut off aid for Ukraine, & Keep Guns in the wrong hands. Please.
Lack of Semantic Correlation	Anyone have a fix on the amount of our money that has been sent out for laundry in Ukraine?
Use of Satire	Let's fortify schools - we have billion for Ukraine.
Lack of Specificity	Let's not forget about the million Hunter got from Ukraine while Kushner got billion with a B from Saudi Arabia. I've rebranded this new Republican party as the Christian Nationalist Hypocrisy Party.

5. Findings

In this section, we answer the three research questions by qualitative analysis of predictions performed by GPT-4 Few Shot on the entire collection of tweets.

5.1. Geographical Sentiment Trends (RQ1)

To answer RQ1, we performed an exploratory data analysis to determine the volume and the relative volume of the tweets per state, calculated as a ratio of volume over state population. Unsurprisingly, for volumetric analysis, states with the highest populations had the highest amounts of tweets. For percentage analysis, West Virginia and Nevada, two states perceived as relatively isolationist, had the highest ratio of tweets to state population. Once the tweets were categorized, those labeled as neutral or inconclusive were disregarded for analysis purposes. The remaining tweets were grouped to calculate a normalized net sentiment (NNS) rating of the state using the difference of polarities (opposing *minus* supporting) divided by the total number of tweets for the state. The lower the NNS rating, the higher the portion of tweets that opposed aid to Ukraine. Indiana had the most negative NNS of -0.8375, while Vermont had the most positive NNS of +0.3333. Of states with a larger population, Virginia had an NNS of -0.02387, indicating a significantly higher portion of tweets favoring aid.

In Figure 3, the bubble size for each state represents the relative volume of total tweets analyzed (supporting and opposing tweets), and the color represents the NNS rating on a sliding colormap scale where a -1 rating corresponds to red, and a +1 rating corresponds to green, with the intensity of the color signifying the strength of the sentiment. As seen in the figure, almost all the states show a net negative sentiment, with the exception of the state of Virginia (including the Washington D.C. area), which is slightly positive in net sentiment. It should be noted that the sentiment analysis is limited to this dataset only, which was scraped from Twitter and may not be an

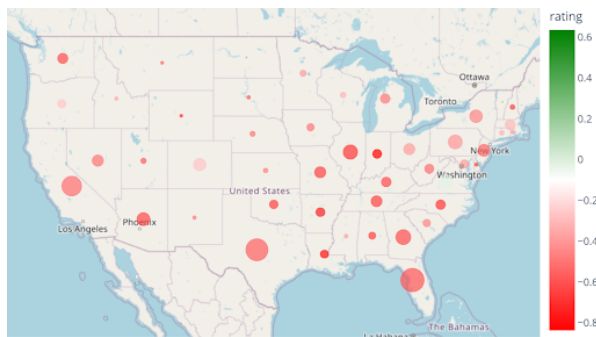


Figure 3: Normalized Net Sentiment by State.

exhaustive collection of tweets on the topic.

5.2. Common Themes for Supporting/Opposing Aid (RQ2)

The most common theme that we encountered in favor of the U.S. Aid (see Figure 4) was *Humanitarian Reasons*, using terms like “medical aid” and “supplies.” This theme supports general aid to help Ukraine as opposed to a particular legislative package (e.g., They have not forgotten. They are doing more to assist Ukraine than most countries. Medical aid, supplies, food, ambulances, and taking in refugees!). The second most common theme was *Supports Aid-related Legislation* (e.g., She was good at getting the squad in line to vote yes on sending money to Ukraine). In the above example, Squad refers to a group of progressive democrats that typically oppose excessive military aid and related legislation.

The next three frequent themes are: *Highlights Opposition to Russian Aggression* (e.g., At this point, the West cannot allow Russia to take control of East Ukraine & t/ corridor to the Baltic Sea. IF SO Russia would control all food attempting to leave the Bread Basket of Ukraine which feeds millions, if not billions of ppl, around t/ world.), *Anti-GOP/Pro Democrats* (e.g., Make no mistake. The Republican party wants to end aid to Ukraine and let Russia illegally “annex” the entire country. Real American patriots remember Russia said they might want to take back Alaska too. American patriots are voting to stop Russia. We are angry and we are voting.), and *Anti-Trump* (e.g., Had Trump’s insurrection succeeded there is no doubt in my mind the U.S. would have been included on this list. GOP doubles down on outright hostility towards Ukraine. If they take Congress in Nov will aid cease?).

We also identified the major themes opposing U.S. Aid to Ukraine (see Figure 5). These include *Alternative Areas for Spending* (e.g., We don’t have money for infrastructure, education, or healthcare, but

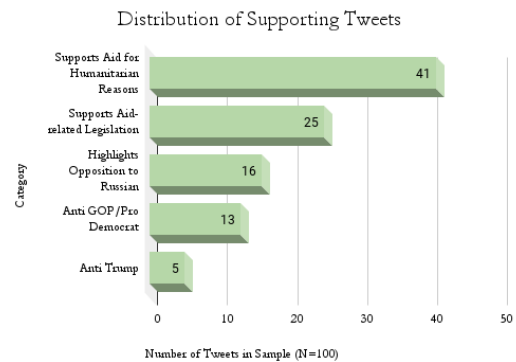


Figure 4: Themes Supporting U.S. Aid to Ukraine.

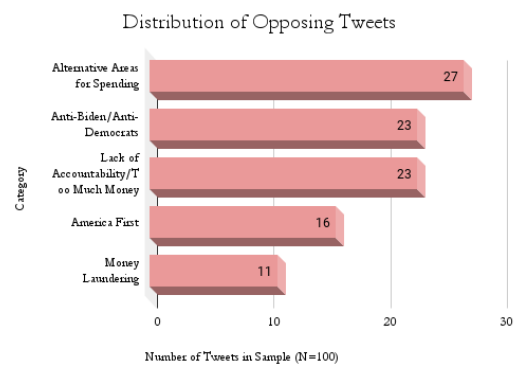


Figure 5: Themes Opposing U.S. Aid to Ukraine.

we have enough money to turn Ukraine into a Libya? We have money for war but not the poor got it), *Anti-Biden/Anti-Democrats* (e.g., Every time I see Biden ask more money for Ukraine, I get pissed off again.), *Lack of Accountability/Too Much Money* (e.g., We must require an inspector general to ALL of the that has been sent to Ukraine. There is no record of where the money has gone, and the bombs sent into Poland were from Ukraine. Zsky immediately asked for more money for [truncated].), *America's First* (e.g., The hell with Ukraine, Worry about the citizens of this country first. Biden has created the worst economy that I have ever seen. People in this country can’t afford the essentials and you give billions to other countries. America should always come first!), and *Money Laundering* (e.g., Finance our country, not Ukraine! You and McStain were involved in the corruption going way back. Nothing more than a quasi money laundering scheme!).

5.3. Analysis in Relation to Popular Perception (RQ3)

As indicated by our geospatial analysis, nearly all states displayed a negative net sentiment during the data collection period. However, the popular perception of U.S. aid to Ukraine is slightly different. A series of polls conducted by Pew Research Center⁹ indicates a shifting but slightly positive sentiment during this time period: from 74% who say that the level of support is “Just right” or “Not enough” in March of 2022 to 51% who feel the same in January of 2023. In addition, those who feel that the U.S. is providing “Too much” support rose from 7% to 26% during the same time. There is a divide here between the Republicans and the Democrats as well. Republicans who felt that too much aid was being provided rose from 9% to 40%, while Democrats went from 5% to 15%. However, the demographics of the Twitter/X user base differ significantly from those taking national surveys, which could explain the difference in public opinion.

The difference may also lie in biases found in Twitter/X’s recommender system and how it amplifies certain posts over others. Twitter/X’s system of user communities anchored around influential users is designed to amplify echo chambers, and in a study of 2.7 million tweets, Corsi (2024) found that the system amplified low credibility and high toxicity content. This effect was largely attributable to high-engagement, high-follower tweets, reinforcing the echo chamber effect by putting the voices of the few in the spotlight for many. In addition, Twitter/X’s recommender system has favored the amplification of mainstream political right and right-leaning news sources, though there was no difference for far-right or far-left in amplification (Huszár, 2022).

Moral outrage, anger, or disgust at the perceived violation of a moral norm is a way for humans to signal to others that they are part of a similar group with similar values and can indicate safety to other members of the group. It can also be a measure of social currency, used to signal value as a potential social or romantic partner. The lack of identifiable physical signals on text-based social media, such as Twitter/X, can lead to the over-perception of moral outrage by the observer, leading to increased interpersonal and inter-group hostility in interactions and amplifying echo chambers (Brady et al., 2023). This tendency to experience higher levels of moral outrage than intended could also contribute to increased attribution of negative emotions to tweets that would otherwise be considered

slightly negative or even neutral, possibly increasing their capacity to go viral (Schöne et al., 2021) and at a faster pace (Antypas et al., 2023).

We must also consider the potential effects of outside interference on our data and subsequent analysis. Bot networks are now considered increasingly fundamental tools of the political machine and warfare (Woolley, 2016). Shen (2023) identified 13.4% accounts producing 16.7% of tweets relative to the Ukraine conflict as bot accounts, a significant portion of the conversation. In addition, Additionally, on October 28, 2022, Elon Musk purchased Twitter and immediately began making significant changes to the policies and staffing of the organization, which could have impacted the user demographics significantly.

6. Conclusion

Since the onset of the Russian-Ukraine conflict, the United States has countered Russian aggression in Eastern Europe through financial support. While the aid allows the U.S. to support humanitarian efforts and maintain regional stability, it divides the public opinion on the necessity of such spending on foreign conflicts. In this paper, we collect tweets related to the U.S. aid to Ukraine and analyze the public opinion through a mixed methods approach. Our biggest contribution is in terms of methodology, as we explore a novel approach to opinion mining using minimal training data. We also compare various approaches to analyze Twitter sentiment.

We answer three research questions. RQ1 investigates how public opinion on the topic of U.S. aid to Ukraine varies across the states. We select tweets belonging to the continental U.S. and match the coordinates to corresponding states. We calculate the normalized net sentiment (NNS) rating for each state, which reveals that Indiana has the most negative NNS of -0.8375, while Vermont has the most positive NNS of +0.3333. RQ2 looks at the major arguments — emerging from a thematic analysis — supporting or opposing U.S. aid to Ukraine. Humanitarian reasons are the most common theme for tweets favoring aid, while the most common themes against the aid highlight alternative areas of funding for domestic issues and the potential misuse of funds due to corruption. RQ3 explores why our data reflects a negative sentiment towards the aid while the general opinion for the time frame, while declining over time, was generally in support. Possible reasons include Twitter’s recommender system and its potential biases, the changing demographics of Twitter users, and the human tendency to interact more frequently with content

⁹<https://www.pewresearch.org/short-reads/2023/02/23/what-public-opinion-surveys-found-in-the-first-year-of-the-war-in-ukraine/>

containing negative sentiment.

7. Limitations and Future Work

There are some limitations to this work. The data was scraped from Twitter and is a subset of tweets on the topic. Therefore, our dataset (containing 11,289 tweets) is not an exhaustive representation of tweets on the topic, and the public opinion assessed is specific to this dataset only. We aim to predict the tweet sentiments as accurately as possible in a sample-efficient situation (limited annotation). While GPT-4 outperforms other models, we do not perform any statistical significance tests or compare the performance with other emerging open-sourced LLMs. Also, Twitter/X limits the posts to 280 characters. Due to the conciseness of the tweets, the underlying sentiment is not always articulated clearly and makes sense only when the user's background and context are known. The interaction of human psycho-social reactions to social media content and how recommender systems intersect to decide what content will become popular is still a nascent field of study. In addition, at this time, there is no requirement for systems to address bias in their recommendations. In the future, we would like to analyze a larger sample of tweets for themes, along with comments, to better understand the context involved and the biases in the recommended feed.

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